



ZECOMERCE

Developed By

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1. Introduction

1. 1 Project Profile

Introduction

ZEcommerce is an Android-based e-commerce application with Admin and User modules.

It allows Admins to manage products, stock, offers, and orders while Users can browse products, place orders, manage wallet, and view invoices. The system uses Firebase Cloud Firestore for real-time data management, and Kotlin + XML for Android app development.

Objective

The objective of ZEcommerce Project is to automate:

- Product management through Admin application
- Product browsing and shopping through User application
- Cart and order management system
- Offer and discount management by Admin
- Wallet balance and transaction management
- Order tracking and cancellation for users
- Invoice generation and viewing system
- User profile management and updates
- Admin order monitoring and product control

System Overview

ZEcommerce is divided into two separate applications with different user roles:

1. **Admin Application**
2. **User Application**

Each application has specific responsibilities and access to particular modules. The Admin application is used to manage products, stock, offers, and customer orders. The User application allows customers to browse products, search and filter items, add products to the cart, place orders, and view invoices.

The system is designed to be user-friendly, secure, and scalable.

System Rolls :

System Rolls
• User
• Admin

2. Environment Description

2.1 Technology Used

Technology	Description
Android Studio	Integrated Development Environment (IDE) used to develop Android applications
Kotlin	Programming language used to implement application logic
XML	Used for designing the user interface layouts
FireBase Authentication	Used for secure user Login , Registration And Session Management
Cloud Firestore	Cloud-based NoSQL database used for storing application data
Supabase	Cloud-based Storage Used for storing images
Gradle	Build automation tool used in Android development

2.2 Hardware Requirement

Component	Requirement
Processor	Intel Core i5 or higher
RAM	Minimum 4 GB
Storage	Minimum 20 GB free disk space
Device	Laptop or Desktop Computer
Testing Device	Android Smartphone

2.3 Software Requirement

Component	Software
OS (Operating System)	Windows 11
Software	Android Studio
Browser	Google Chrome , Microsoft Edge

3. System Requirement And Specification

3.1 Data Dictionary

PRODUCTS (Collection)		
Field	DataType	Description
• ProductId	String	Unique Product ID
• Product Name	String	Name Of The Product
• Description	String	Product Description
• Price	Number	Product Price
• Stock	Number	Product Stock
• Category	String	Product Category
• ImageUrl	String	Product Image URL
• Created At	TimeStamp	Product Creation Time
• Active	Boolean	Soft Delete Product

Offers (Collection)		
Field	DataType	Description
• Offer Id	String	Unique Offer ID
• Title	String	Offer Title
• Active	Boolean	Offer Is Active Or InActive
• ImageUrl	Number	Offer Banner Image Url
• Products	Collection	Inside Offer Products Collection
➤ Product Id	String	Related Products Product Id
➤ Original Price	Number	Product MRP Price
➤ Offer Price	Number	Product Selling Price
➤ Product Name	String	Name Of The Product
➤ Product Image Url	String	Product Image Url

Reviews (Collection)		
Field	DataType	Description
• ReviewId	String	Unique Review ID
• UserId	String	User Id Who Posted Review
• ProductId	String	Related Product
• UserName	Number	Username Who Posted Review
• Rating	Number	Review Rating
• Message	String	Review Message/Comment
• Created At	TimeStamp	Product Creation Time
• Likes	Number	Total Likes On Particular Review

Cart (Collection)		
Field	DataType	Description
• CartId	String	Unique Cart ID
• UserId	String	User Id For Cart
• ProductId	String	Related Product
• Quantity	Number	Product Quantity
• Original Price	Number	Product MRP Price
• Price	Number	Product Selling Price
• IsOffer	Boolean	Any Offer On Product
• ProductName	String	Name Of The Product
• ImageUrl	String	Product Image Url

USER DETAIL (Collection)		
Field	Data Type	Description
• UserId	String	Unique user ID from Firebase Authentication
• Full Name	String	Name Of The User
• Email	String	User Email Address
• Phone	String	Contact Number
• Balance	Number	User wallet balance
• ProfileImageUrl	String	User Profile Picture Image URL
• Created At	TimeStamp	Account Creation Time

Orders (Collection)		
Field	Data Type	Description
• Order Id	String	Unique Order ID
• User Id	String	User Id Who Place The Order
• Customer Name	String	Name Of The User
• Receiver Name	String	Receiver Name
• Email Id	String	Email Id For Contact
• Phone Number	String	Phone Number
• Address	String	Location Where Order Delivered
• Payment Mode	String	Method Like UPI /Token
• Payment Status	String	Success Or Pending
• Total Amount	Number	Total Amount Of Order
• Order Status	String	Order Status Placed, Cancelled,Completed.
• Created At	Time Stamp	When Order Placed
• Items	Collection	Ordered Items
➤ Items Id	String	Unique Item Id
➤ Product Id	String	Related Products Product Id
➤ Image Url	String	Image Url Of The Product
➤ Product Name	String	Name Of the Product
➤ Is Offer	String	Any Offer On Product
➤ Original Price	Number	Product MRP Price
➤ Price	Number	Product Selling Price
➤ Quantity	Number	Item Quantity

3.2 Feasibility Analysis

Feasibility analysis is essential in determining the viability of developing any mobile Application. This analysis evaluates economic, technical, and behavioral factors to ensure the project's success.

3.2.1 Economic Feasibility

Economic feasibility assesses whether the project is financially viable and cost effective. The development of the ZECOMERCE Android application involves certain costs related to development tools, cloud services, and maintenance. The primary cost components include:

- **Development Costs:** Expenses related to the development process using Android Studio and the implementation of application features.
- **Cloud Services Costs :** Expenses associated with Firebase And Supabase services such as Firebase Authentication, Cloud Firestore database, and Supabase Media Storage.
- **Maintenance Costs:** Regular updates, bug fixes, performance improvements, and security enhancements. security patches, and technical support.
- **Testing Costs :** Testing the application on different Android devices and ensuring smooth performance.

Since most of the tools used for development such as Android Studio , Supabase and Firebase free-tier services are available at minimal or no cost, the overall project development cost is low. Therefore, the ZECOMERCE application is economically feasible.

3.2.2 Technical Feasibility

Technical feasibility evaluates the technology requirements and the ability to implement the project successfully. The following aspects are considered:

- **Technology Stack:** The application is developed using Kotlin programming language and XML layouts in Android Studio for building the Android user interface.
- **Database Infrastructure :** The system uses Firebase Cloud Firestore, a NoSQL cloud database, Supabase cloud Media Storage that supports real-time data synchronization and scalability.
- **Authentication System:** Firebase Authentication is used to provide secure user login, registration And Secure Session Management functionality.
- **Development Complexity:** The project requires expertise in full-stack Application development, but with proper planning and execution, it is feasible.

The availability of required technologies and development tools ensures that the ZECOMERCE Android application is technically feasible

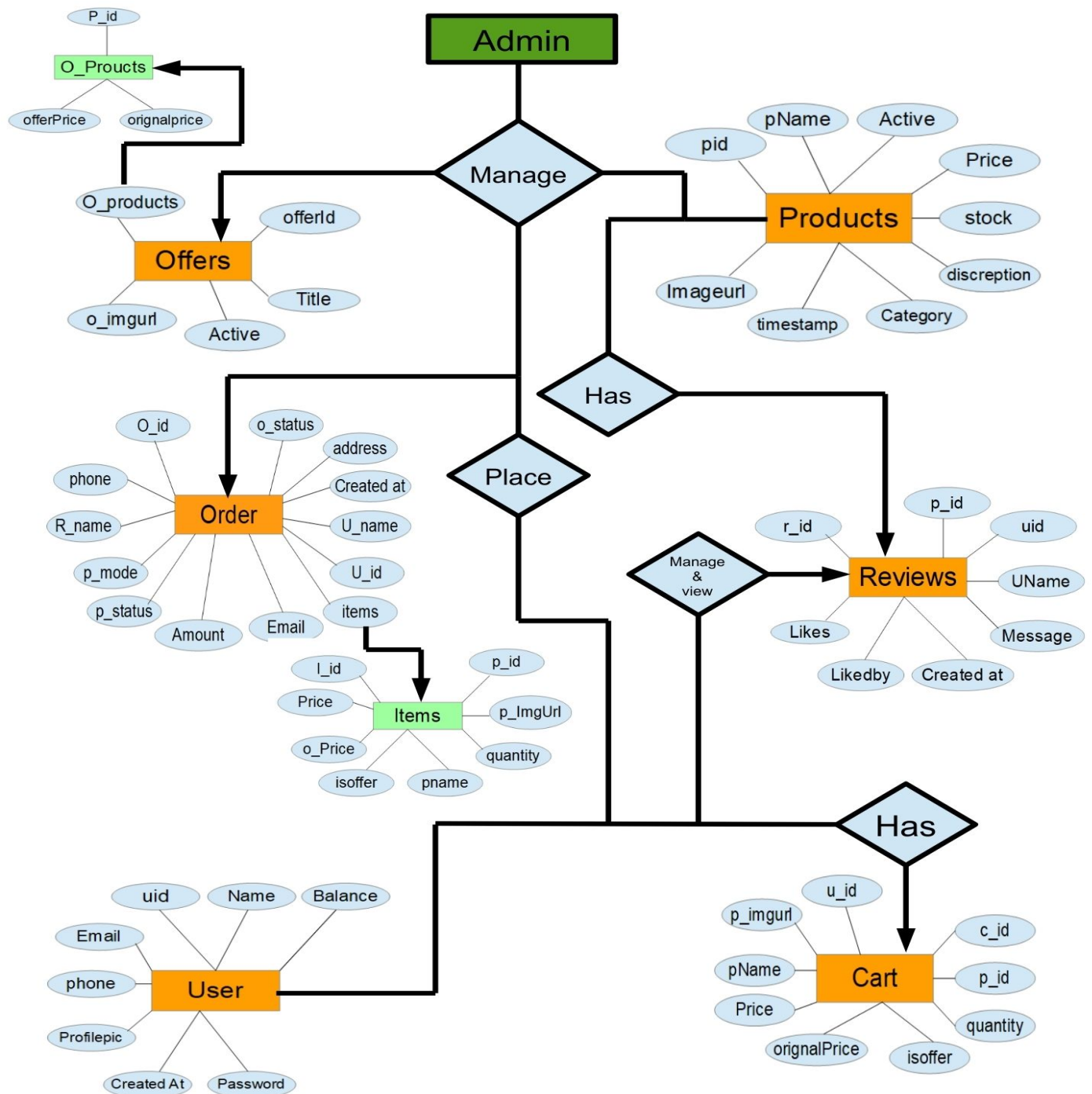
3.2.3 Behavioral Feasibility

Behavioral feasibility focuses on user acceptance, adaptability, and overall demand for the application. The following factors are considered:

- **User Demand:** With the increasing trend of online shopping, users prefer mobile applications for purchasing products conveniently
- **User-Friendly Interface:** The application provides a simple and intuitive user interface that allows users to easily browse products, place orders, and manage their accounts
- **Customer Trust & Security:** The use of Firebase Authentication and secure database services ensures user data protection and reliability.
- **Admin Control :** The separate Admin application helps administrators manage products, offers, and orders efficiently

The positive acceptance of mobile shopping applications by users indicates high behavioral feasibility for the ZECOMERCE system.

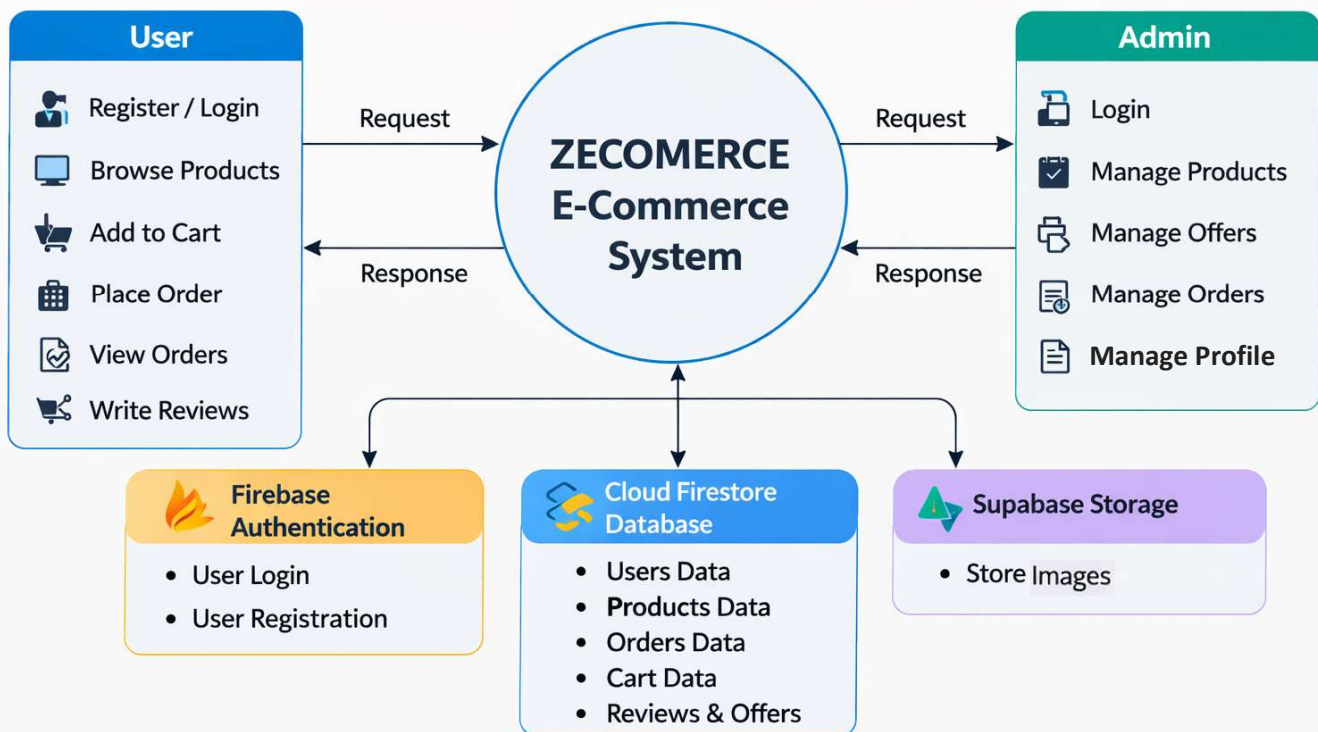
3.3 E R Diagram



3.4 Data Flow Diagram

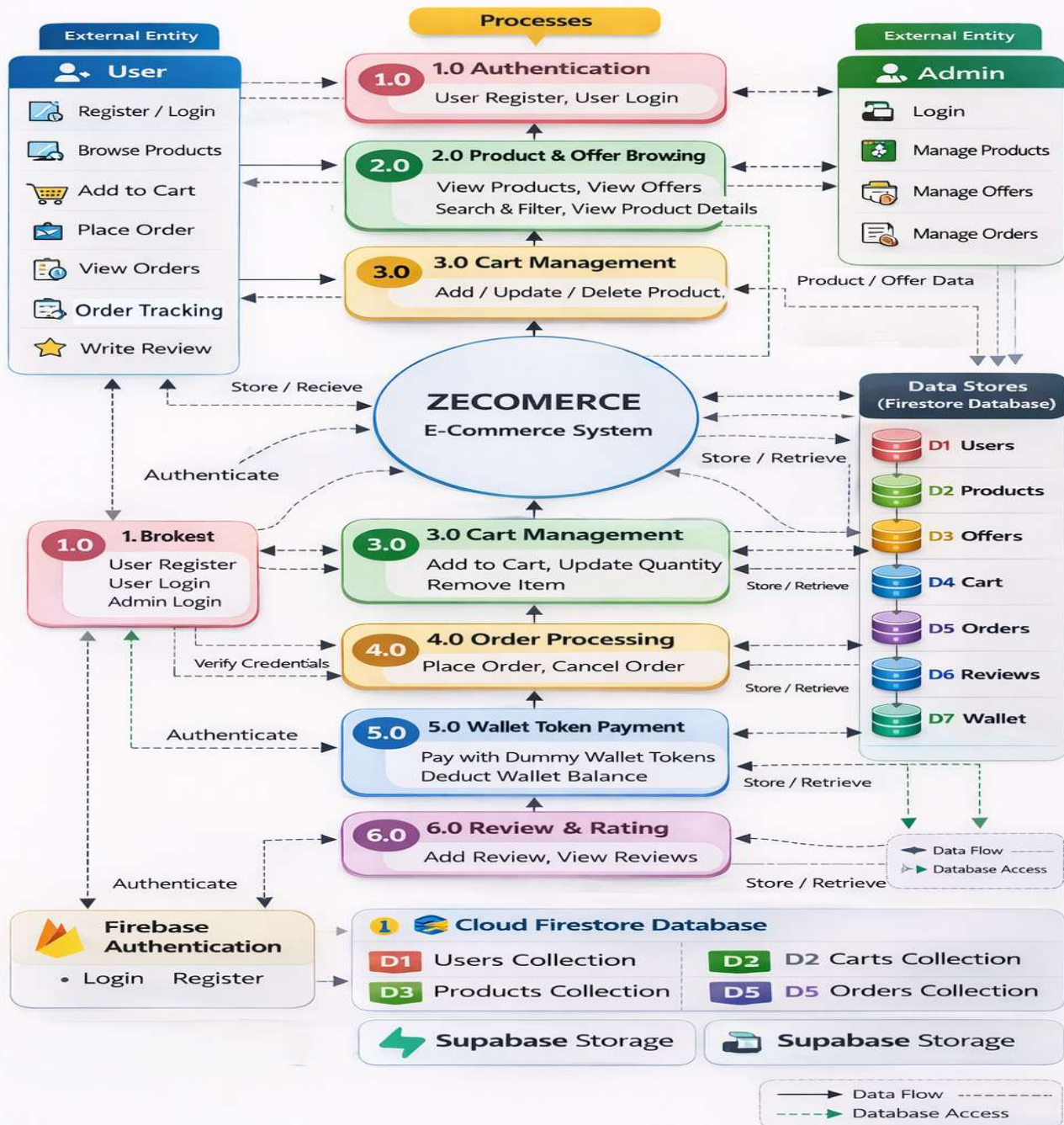
DFD Level 0

DFD Level 0 – ZECOMMERCE (E-Commerce Application)



DFD Level 1

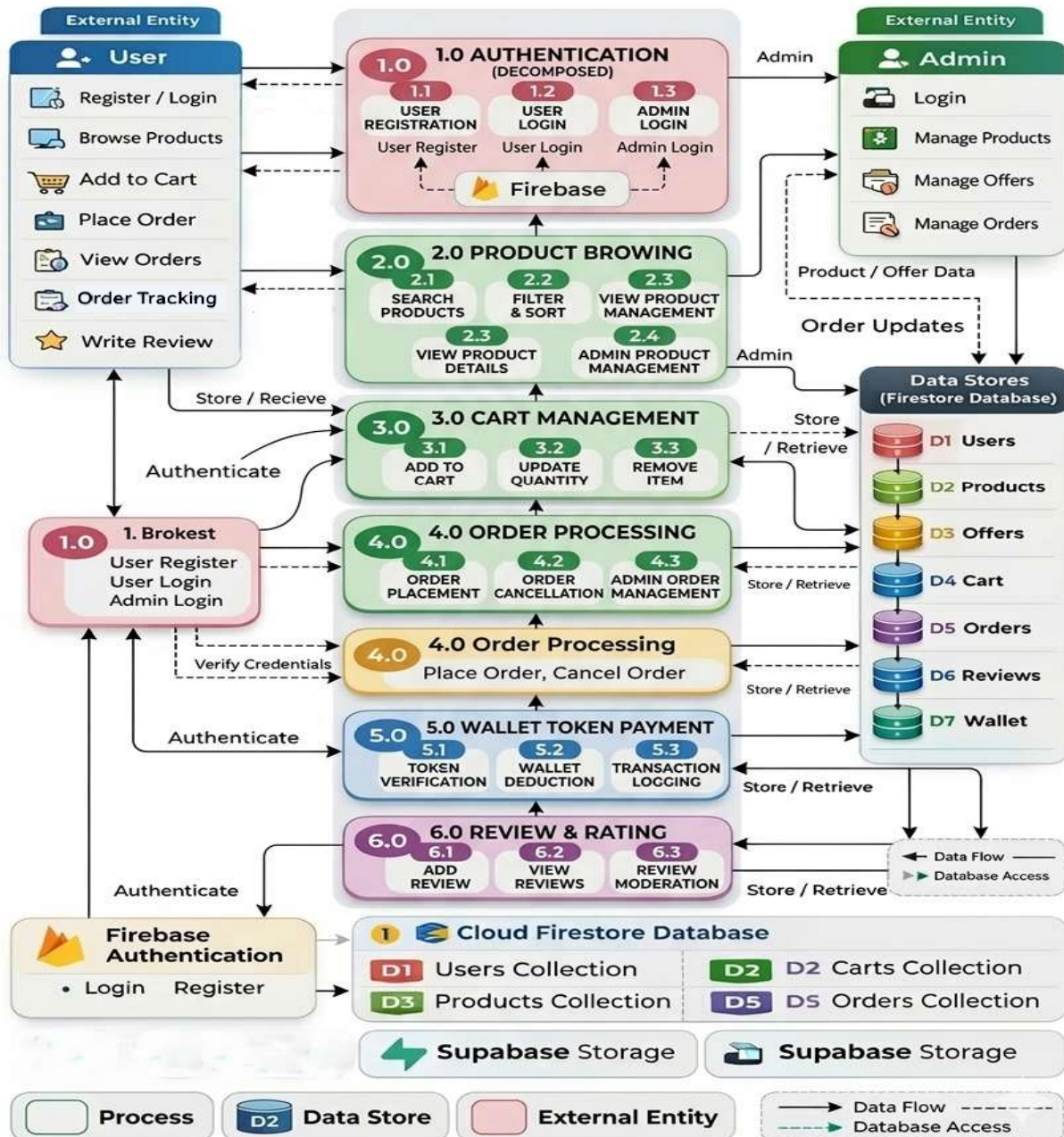
DFD Level 1 – ZECOMMERCE (E-Commerce Application)



DFD Level 2

DFD Level 2 – ZECOMMERCE (E-Commerce Application)

DETAILED LEVEL 2 VIEW OF SELECTED PROCESSES



User Role & Functionality

1. Admin

The Admin has full control over the system through the Admin Application. The Admin is responsible for managing products, offers, and customer orders.

Admin can perform the following functions:

- Add new products to the system
- Update product details such as price, stock, and description, Delete products
- Create offers , Add products to offers and manage offer prices
- View and manage all customer orders
- Update order status (Placed, Completed, Reject)
- Manage admin profile information

2. User

The User interacts with the system through the User Application to browse and purchase products.

Users can perform the following functions:

- Register and login using Firebase Authentication
- Browse and search products
- View product details and available offers
- Add products to cart, Manage cart items and quantity
- Place orders for selected products, View order history and order details
- Cancel orders if the status is Pending or Placed
- Write reviews and rate products
- Manage personal profile and wallet balance

3. Customer Interaction

This functionality focuses on user engagement and feedback within the application.

Users can:

- Write reviews for purchased products
- View ratings and feedback from other users
- Share product experience through reviews

This feature helps improve product quality and builds trust among users.

4. Order and Cart Interaction

This functionality manages the shopping process for users.

Users Can :

- Add products to the cart
- Modify cart items and quantities
- Remove products from the cart
- Place orders securely
- Track order status and history

4. Software Project Planing

4.1 Scope

The ZECOMERCE Android Application is an E-Commerce platform designed to provide a convenient way for users to browse products and place orders through a mobile application. The system consists of two separate applications: Admin Application and User Application.

The Admin can manage products, offers, orders, and their own profile through the Admin application. Admin has full control over product listings, offer creation, and order management.

The User can register and login using Firebase Authentication, browse and search products, add products to the cart, place orders, write product reviews, and manage their own profile and wallet balance.

The system is designed to be user-friendly, secure, and efficient, providing a smooth shopping experience while ensuring proper management of products and orders.

4.2 Objective

The objective of the ZECOMERCE application is to automate the online shopping process and simplify product management through mobile applications.

The main objectives include:

- Product management by Admin
- Offer and discount management
- Online product browsing and searching
- Shopping cart management
- Online order placement and order tracking
- User profile management
- Product review and rating system
- Secure user authentication using Firebase
- Efficient product and order data storage using Firebase Firestore

4.3 Expected Advantages

- Reduces manual errors
- Increases transparency
- Provides a convenient online shopping experience
- Easy product browsing and ordering through mobile application
- Secure user authentication using Firebase Authentication
- Centralized data storage
- Real-time updates

Module Description

1 Product Management

This module allows the Admin to manage all product-related information.

- Add new products
- Update, Delete products
- View all product listings
- Manage product categories and stock

2 Offers Management

This module allows the Admin to create and manage promotional offers.

- Create new offers
- Add products to offers
- Update offer details
- Activate or deactivate offers
- Users can view discounted products available in offers.

3 Cart Management

The cart module allows users to manage products before placing an order

- Add products to cart.
- Update product quantity.
- Remove products from cart.
- View total cart amount.

4 Review Management

This module allows users to share feedback about products.

Users can:

- Write product reviews
- Rate products
- View other users' reviews

This helps other customers make better purchasing decisions.

5 Order Management

This module manages customer orders.

Admin can:

- View all orders
- Update order status (Placed, Completed, Reject)
- Manage order processing

Users can:

- Place new orders
- View their order history
- Cancel orders if the status is Pending or Placed

6 Profile Management

Users can manage their personal account information.

- View and update profile details
- Manage wallet balance
- View account information

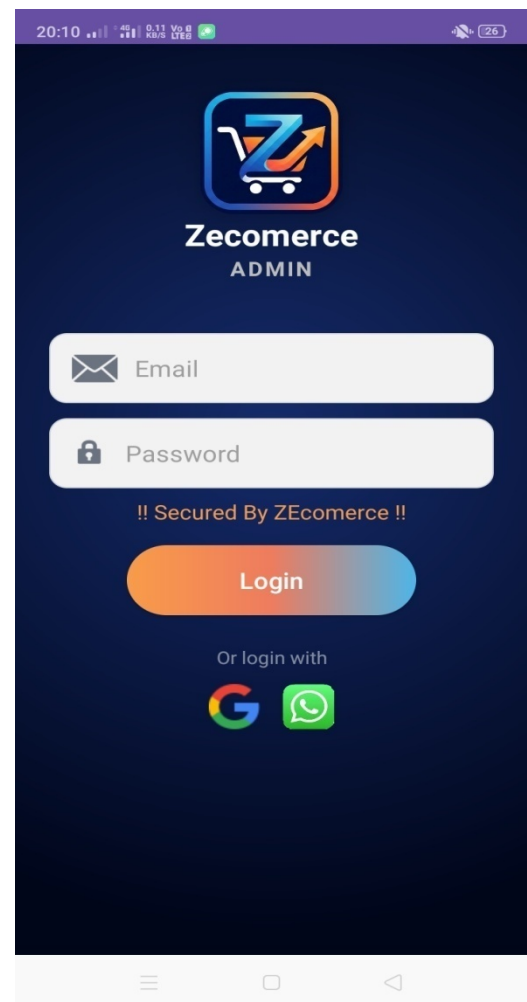
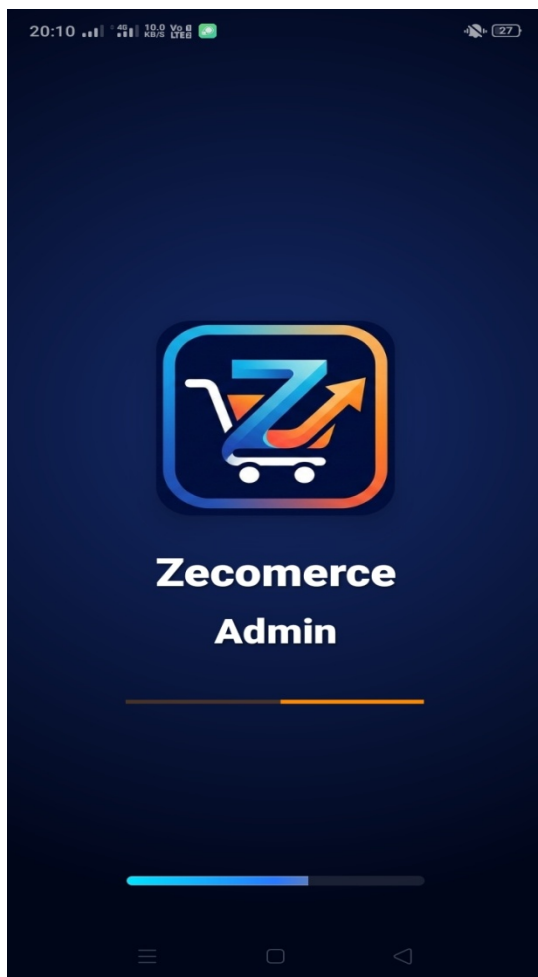
Admin can also manage their own profile in the Admin application

5. System Design

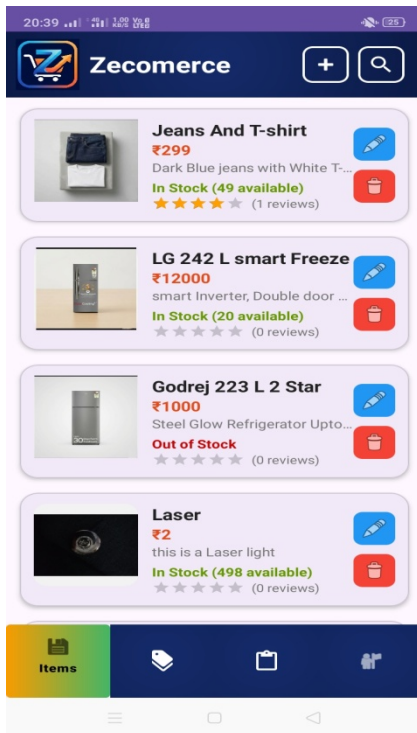
Zecommerce Admin App

Splash Activity

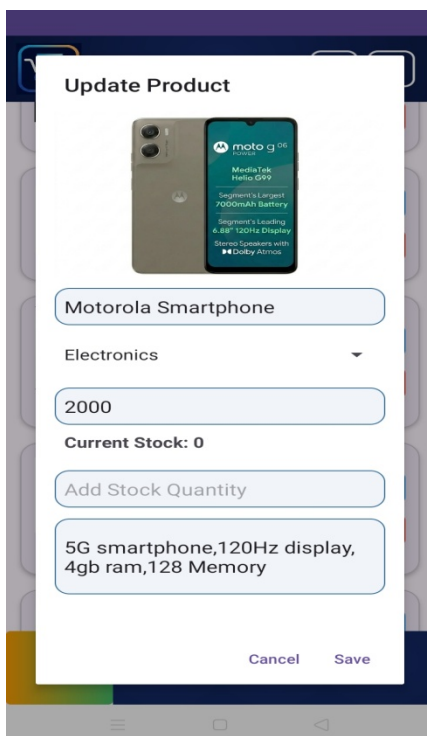
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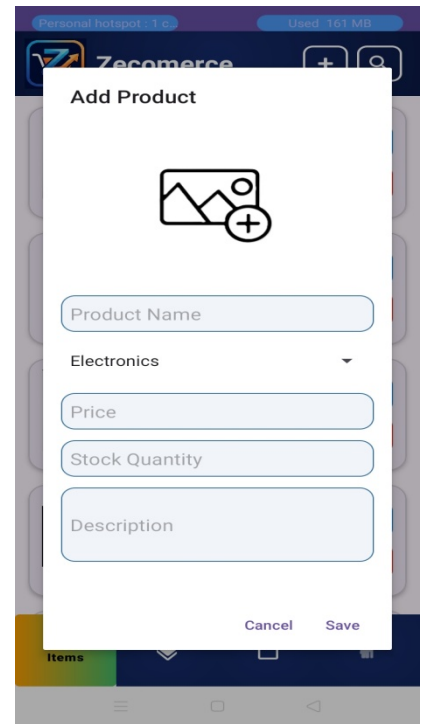
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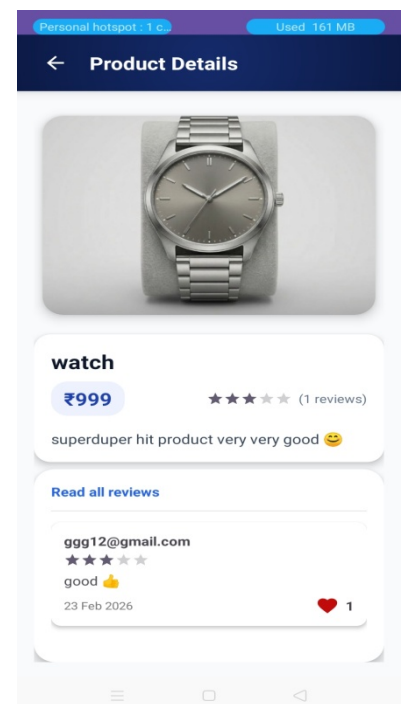
Update Product



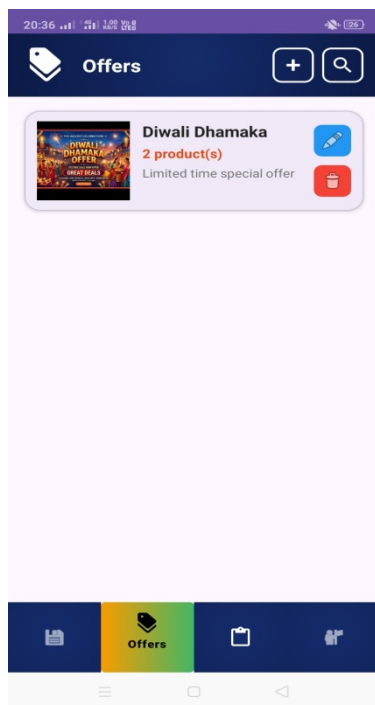
Add Product



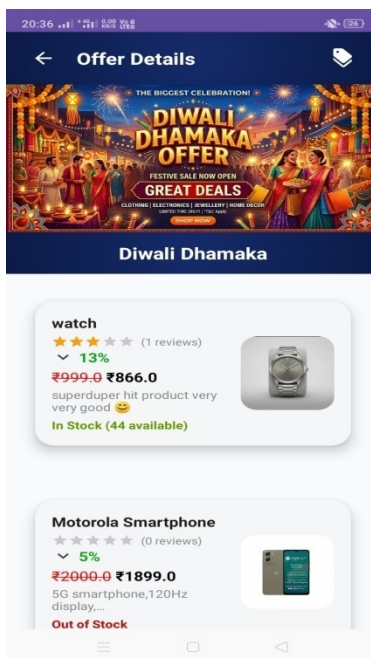
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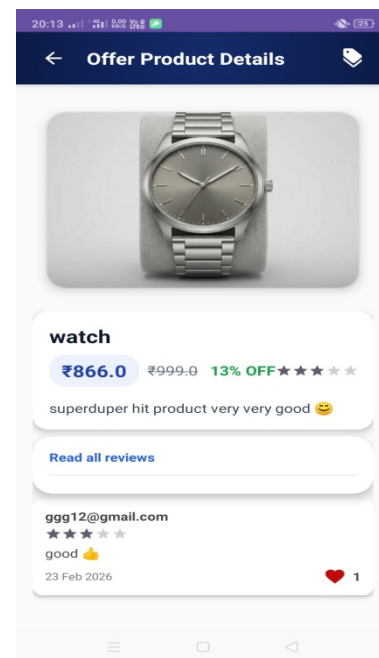
Offers Fragment



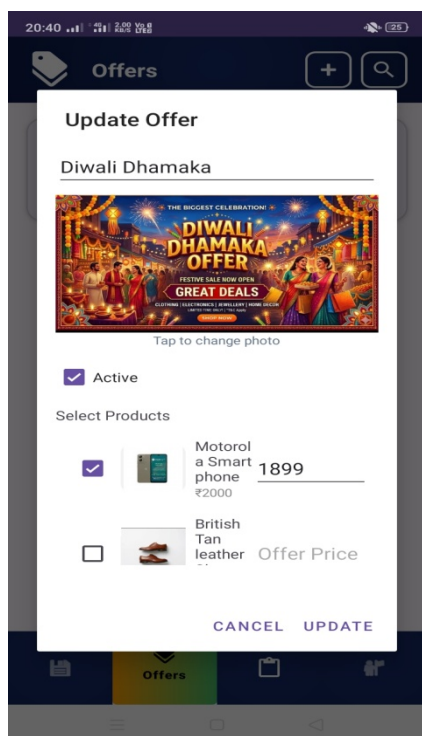
Offer Details



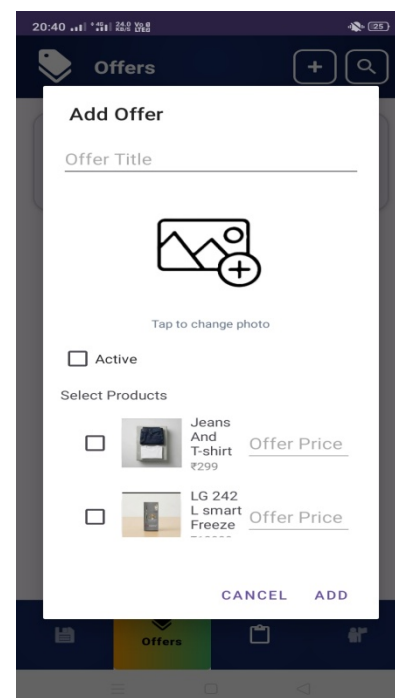
Offer Product Details



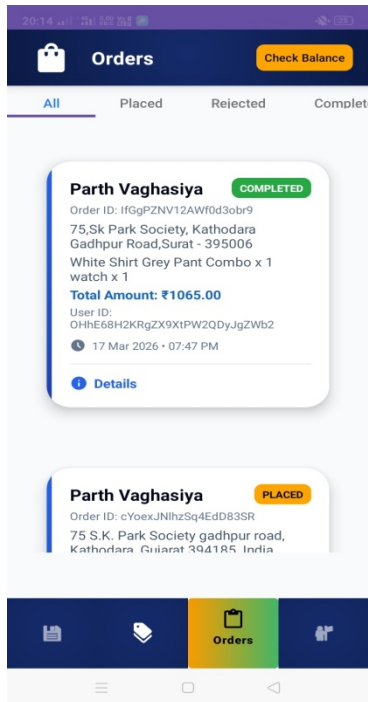
Update Offers



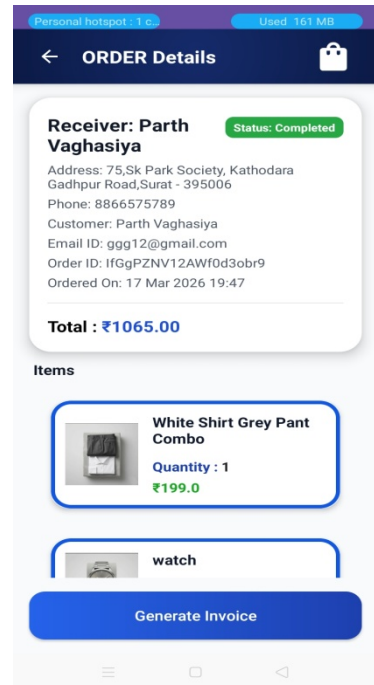
Add Offer



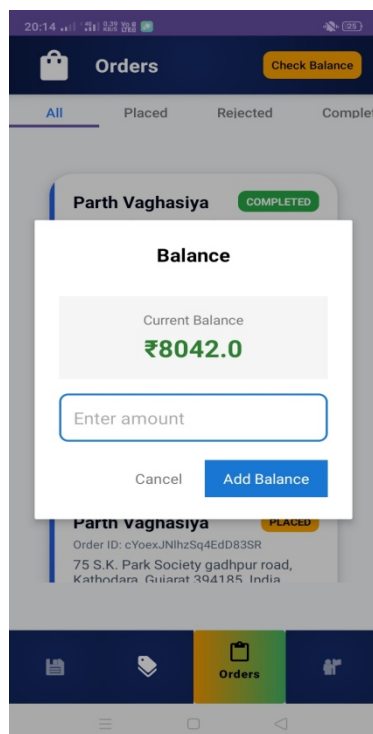
Order Fragment



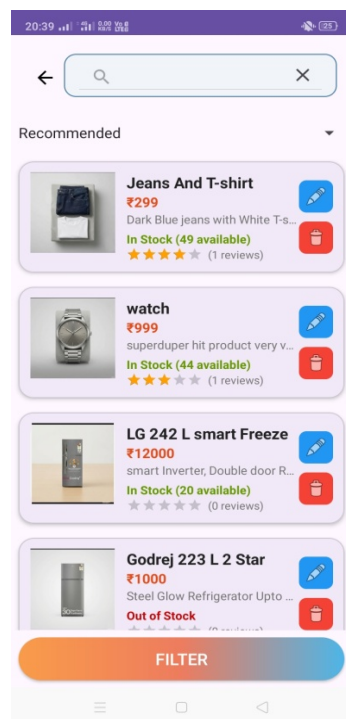
Order Details Activity



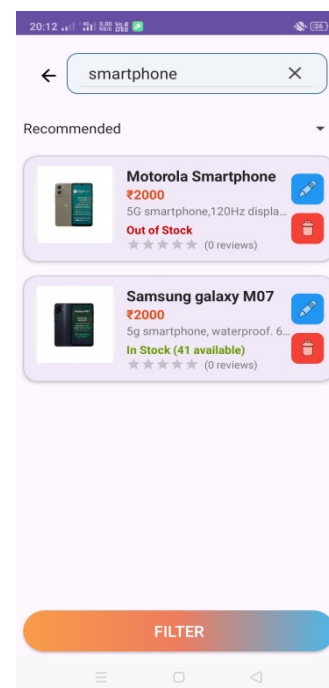
Orders ChekBalance



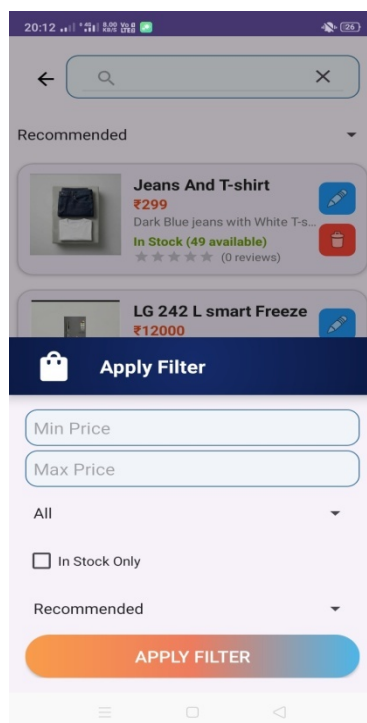
Search Activity

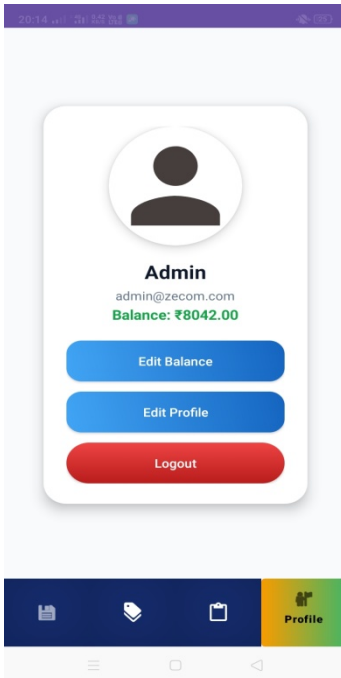
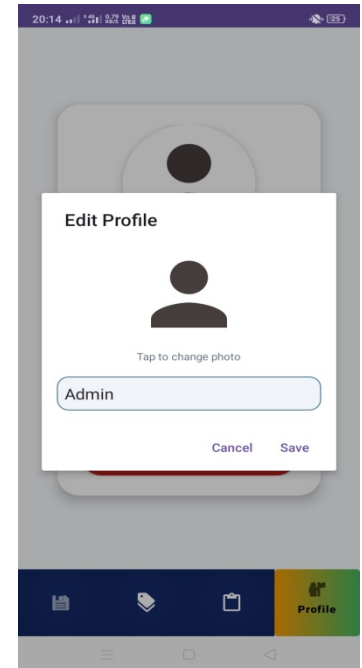
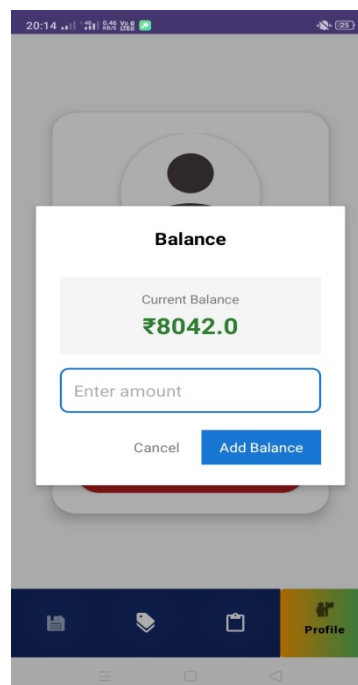


Search Demo



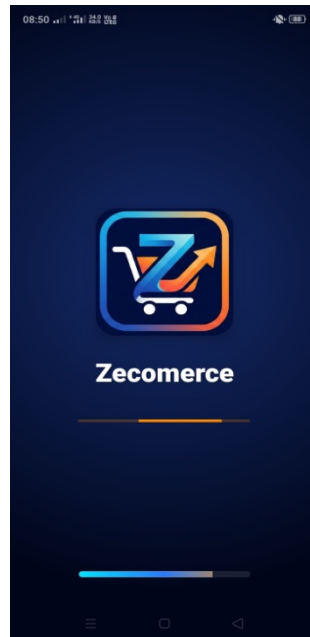
Apply Filter



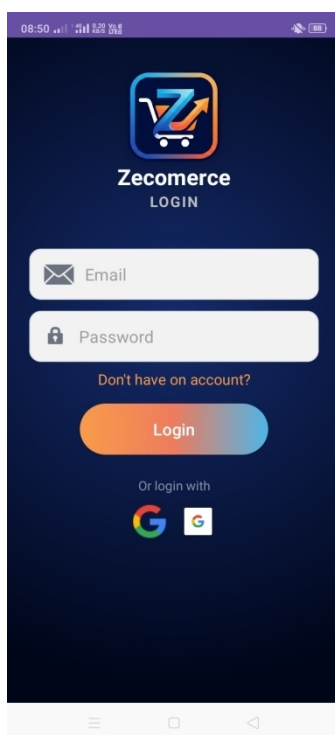
Profile Fragment**Edit Profile****Edit Balance**

Zecommerce User App

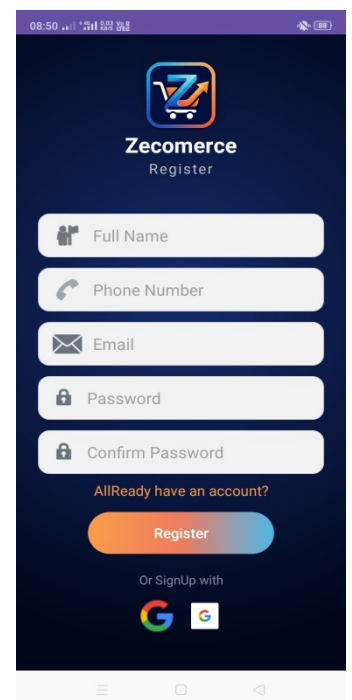
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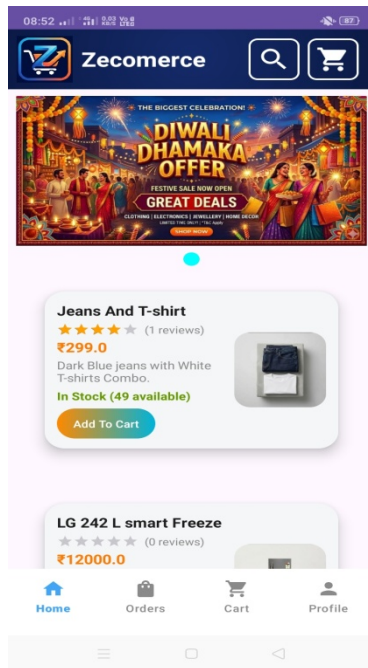
Login Page



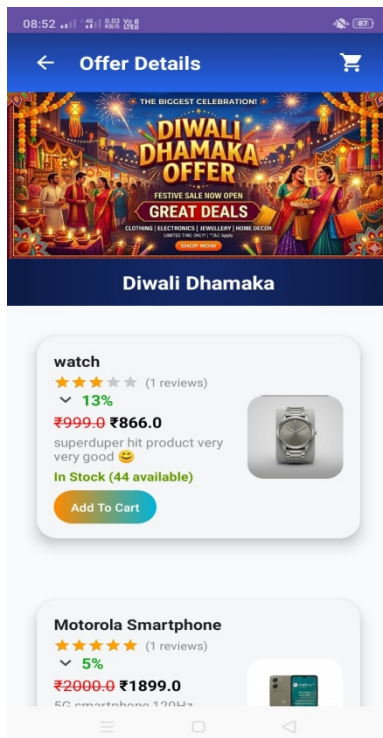
Register Page



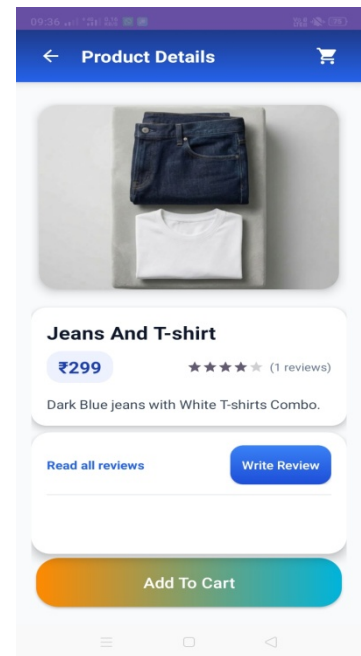
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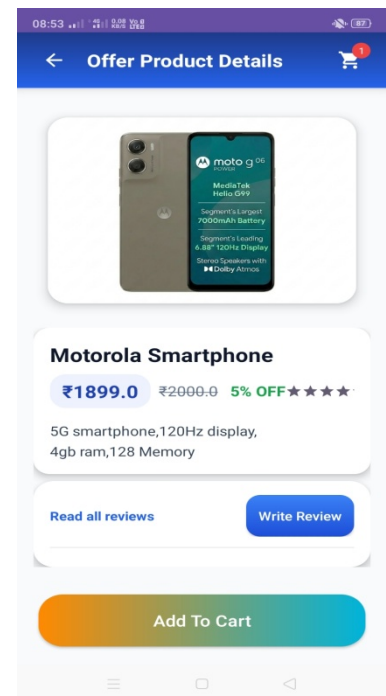
Offer Details Activity



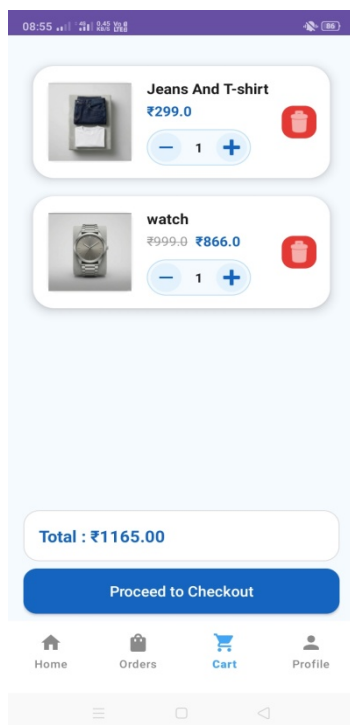
Product Details



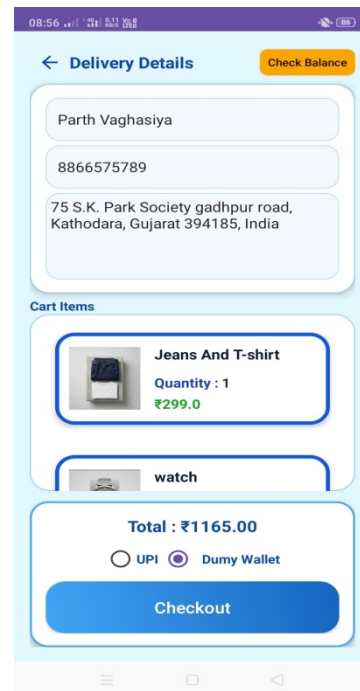
Offer Product Details



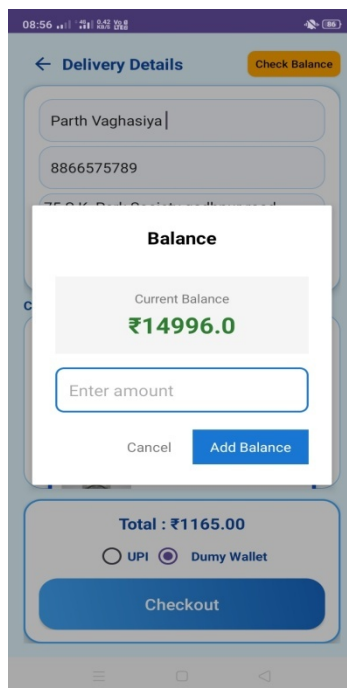
Cart Fragment



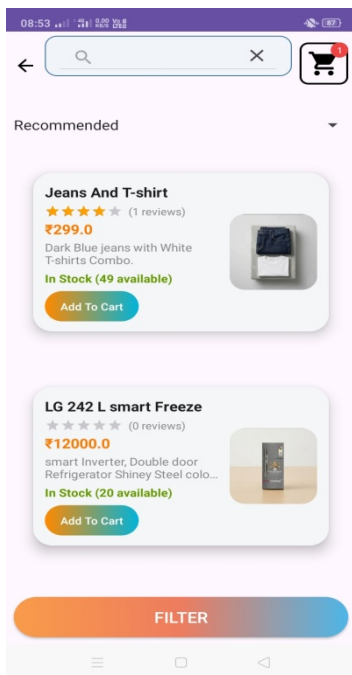
Proceed Order Page



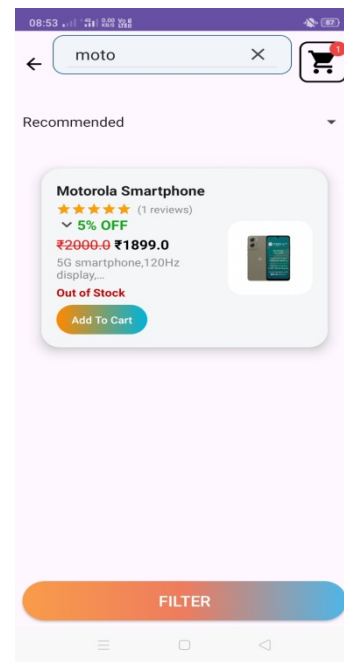
Check Balance



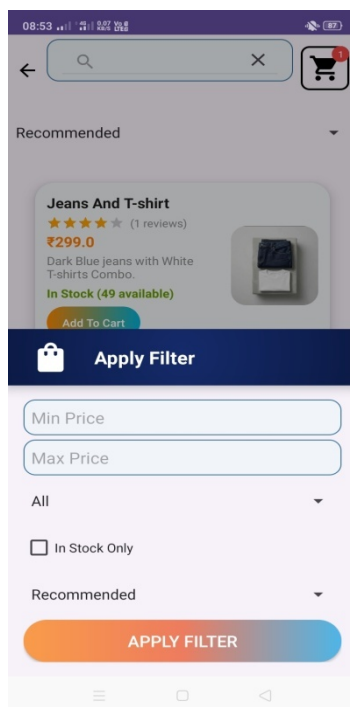
Search Activity



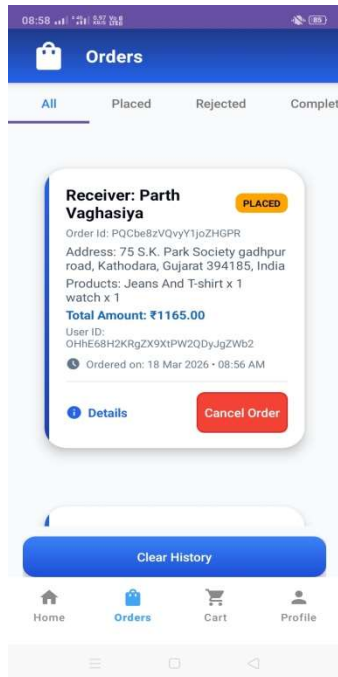
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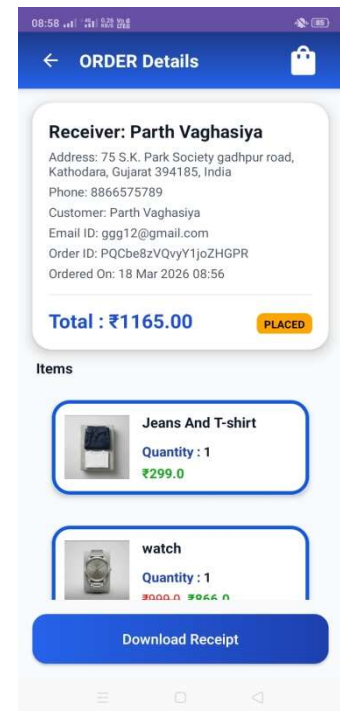
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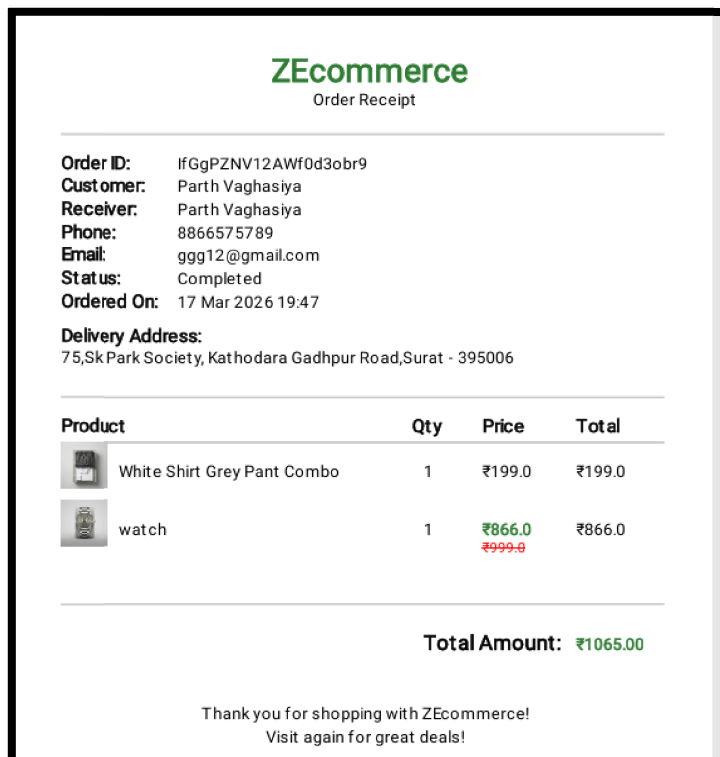
Orders Fragment



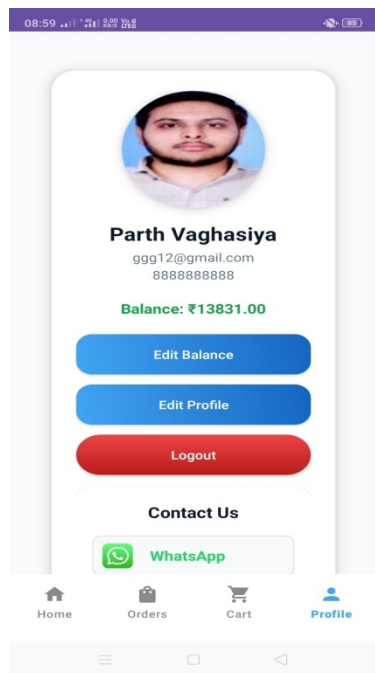
Order Details Activity



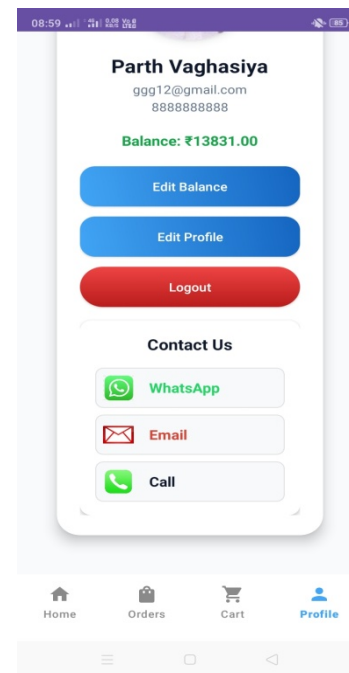
Order Receipt



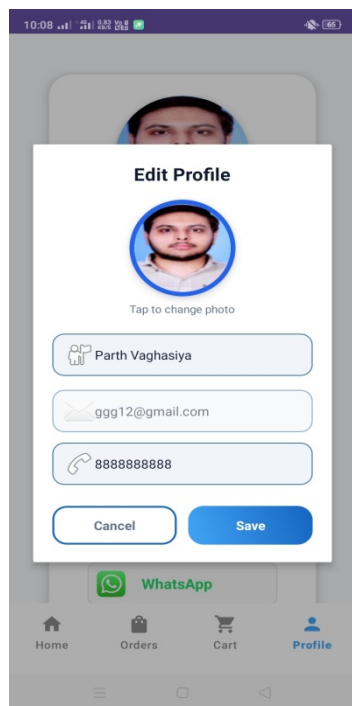
Profile Fragment



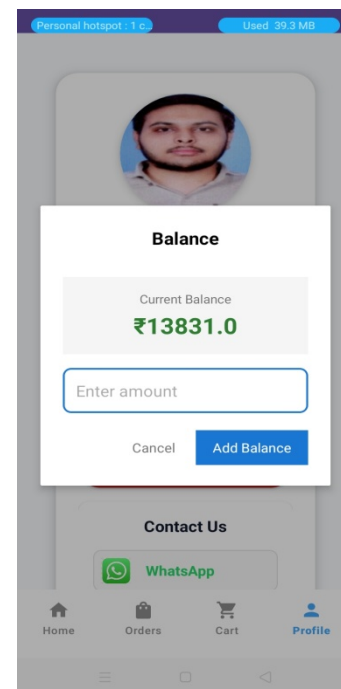
Contact Us



Edit Profile



Edit Balance



6. Software Testing

6.1 Introduction

Software testing is a crucial phase in the software development lifecycle that ensures the reliability, functionality, and security of the any application. The purpose of testing is to identify and resolve defects before deployment, thereby enhancing user experience and system performance. Various testing methodologies, including manual and automated testing, will be employed to verify that all components function as expected.

6.2 Objective

The main objectives of software testing for the any website are:

- **Ensuring Functionality:** Validate that all features, including user authentication, the story creation, the image generation, and other feature works correctly.
- **Enhancing Security:** Identify vulnerabilities and implement security measures to protect user data and financial transactions.
- **Performance Optimization:** Ensure the website performs efficiently under different load conditions.
- **Bug Identification and Resolution:** Detect and fix errors to provide a seamless user experience.
- **Cross-Platform Compatibility:** Test the website across different devices, browsers, and operating systems to ensure accessibility.
- **User Experience Improvement:** Verify usability aspects such as navigation, responsiveness, and ease of use.

Thorough software testing will ensure that any websites meet high-quality standards before its official launch.

6.3 Testing Report

Testing Login & Registration

Test Category	Test Description	Expected Result	Actual Result	Status
Login Screen Access	User can open the login screen from the application start screen	Login screen loads without errors	Login screen opened successfully	Pass
Registration Screen Access	User can open the registration screen from login page	Registration form appears correctly	Registration page displayed successfully	Pass
Form Validation	System validates empty fields and invalid email format	Error message should appear for invalid inputs	Validation messages displayed properly	Pass
Successful Registration	New user registers with valid email and password using Firebase Authentication	User account created successfully and redirected to home screen	User registered successfully	Pass
Duplicate Registration	User tries to register with an email that already exists	System prevents duplicate account creation and shows error	Error message displayed	Pass
Successful Login	User logs in with valid email and password	User redirected to home screen	Login successful	Pass
Invalid Login	User enters incorrect email or password	System displays authentication error message	Proper error message displayed	Pass
Logout Function	User logs out from profile section	User redirected to login screen	Logout successful	Pass

Product Management Testing (Admin)

Test Category	Test Description	Expected Result	Actual Result	Status
Add Product	Admin adds a new product with details	Product saved in Firestore database	Product added successfully	Pass
Update Product	Admin updates product information	Updated information visible in product list	Product updated successfully	Pass
Delete Product	Admin deletes or disables product	Product removed or marked inactive	Product removed successfully	Pass
Product Listing	Admin views list of all products	All products displayed correctly	Product list displayed	Pass
Image Upload	Admin uploads product image	Image stored in Supabase Storage	Image uploaded successfully	Pass

Cart Management Testing

Test Category	Test Description	Expected Result	Actual Result	Status
Add to Cart	User adds product to cart	Product added to cart collection	Product added successfully	Pass
Update Quantity	User changes product quantity	Quantity updated in cart	Quantity updated successfully	Pass
Remove Product	User removes item from cart	Product removed from cart	Product removed successfully	Pass
View Cart	User opens cart screen	Cart items displayed correctly	Cart displayed successfully	Pass
Cart Total	System calculates total cart amount	Correct total price displayed	Total calculated correctly	Pass

Order Management Testing

Test Category	Test Description	Expected Result	Actual Result	Status
Place Order	User places order from cart	Order stored in Firestore	Order placed successfully	Pass
View Orders	User views order history	Order list displayed correctly	Orders displayed successfully	Pass
Cancel Order	User cancels order with pending status	Order status updated to cancelled	Order cancelled successfully	Pass
Admin Order View	Admin views all orders	Admin panel shows all orders	Orders visible to admin	Pass
Update Order Status	Admin updates order status	Order status updated in database	Status updated successfully	Pass

Offer Management Testing

Test Category	Test Description	Expected Result	Actual Result	Status
Create Offer	Admin creates a new promotional offer	Offer should be saved in Firestore database	Offer created successfully	Pass
Add Product to Offer	Admin adds products inside an offer	Product should appear in the offer product list	Product added to offer successfully	Pass
Update Offer	Admin updates offer details	Updated offer information should be saved	Offer updated successfully	Pass
Deactivate Offer	Admin deactivates an existing offer	Offer status should change to inactive	Offer deactivated successfully	Pass
View Offer Products	User views products available in offers	Offer products should display correctly	Offer products displayed successfully	Pass

Review & Rating Testing

Test Category	Test Description	Expected Result	Actual Result	Status
Add Review	User writes a review for a product	Review should be stored in the database	Review added successfully	Pass
Submit Rating	User submits rating for a product	Rating value should be saved correctly	Rating submitted successfully	Pass
View Reviews	Users view reviews under a product	All reviews should be displayed properly	Reviews displayed successfully	Pass
Like Review	User likes a review	Like count should increase	Like recorded successfully	Pass
Review Validation	System checks empty review message	Error message should appear for empty input	Validation message displayed	Pass

Profile Management Testing

Test Category	Test Description	Expected Result	Actual Result	Status
View Profile	User opens the profile screen	User profile information should be displayed	Profile loaded successfully	Pass
Update Profile	User updates personal details	Updated information should be saved in Firestore	Profile updated successfully	Pass
Upload Profile Image	User uploads a profile picture	Image should be stored in Supabase Storage	Image uploaded successfully	Pass
View Balance	User checks wallet balance	Correct wallet balance should be displayed	Wallet balance displayed correctly	Pass
Update Contact Info	User edits phone or email information	Updated contact information should be saved	Contact details updated successfully	Pass

Limitation of System

- Requires stable internet connection
- Dependent on Firebase services
- Limited payment options in current version
- Only available for Android devices
- Requires basic smartphone usage knowledge

Future Scope

- Integration with online payment gateways (UPI, cards, wallets)
- Real-time order tracking system
- AI-based product recommendation system
- IOS application development
- Live customer support chat feature

Conclusion

The ZECOMERCE system makes online shopping simple and efficient by allowing users to browse products, add items to cart, and place orders through an Android application. The system has separate Admin and User roles, which helps in managing products, offers, and orders properly. By using Firebase Authentication and Cloud Firestore, the system provides secure login and reliable data storage. Overall, it improves convenience, reduces manual work, and provides a smooth shopping experience for users.

7. References / Bibliography

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